CS7646 / FINAL EXAM 1 PREPARATION

Learn the rules. Work the numbers. Read the wording.

Machine learning + quantitative finance

A focused, printable companion to your interactive Study Lab. Built around every Exam 1 domain and question group in your supplied pool, with course explanations, worked calculations, retrieval practice, and source-key cautions.

518 questions	308 machine learning + 210 quantitative finance
135 groups	65 ML groups + 70 quantitative finance groups
20 domains	10 ML domains + 10 quantitative finance domains
Pool edition	Rev081020261230; question references use 1-based PDF pages.

Study objective

Explain the mechanism before deciding True or False. For calculations, state the denominator, units, time horizon, information date, and any assumptions.

Prepared from your question pool, the earlier exam review, and lecture materials. It is a teaching synthesis, not a reproduction of the 518 answer keys. Coverage counts describe the supplied pool; they do not predict question frequency on the exam.

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Find the concept you need

Use the PDF bookmarks or the linked chapter names below. Each domain chapter maps every pool group to its controlling rule. Question IDs and pool pages make it easy to return to the original wording.

MACHINE LEARNING	QUANTITATIVE FINANCE
ML-D1 · Learning Paradigms & Problem Framing 38 questions · 6 groups	**QF-D1** · Investment Returns and Compounding 21 questions · 7 groups
ML-D2 · Data & Sampling Discipline 33 questions · 6 groups	**QF-D2** · Risk and Risk-Adjusted Performance 21 questions · 7 groups
ML-D3 · Feature Engineering & Leakage 39 questions · 7 groups	**QF-D3** · Diversification and Correlation 21 questions · 7 groups
ML-D4 · Linear Models & Regression 36 questions · 6 groups	**QF-D4** · Portfolio Optimization 21 questions · 7 groups
ML-D5 · Instance-Based Learning (KNN) 36 questions · 6 groups	**QF-D5** · Market Microstructure and Trade Execution 21 questions · 7 groups
ML-D6 · Decision trees 42 questions · 7 groups	**QF-D6** · Valuation and Mispricing 21 questions · 7 groups
ML-D7 · Ensembles 24 questions · 7 groups	**QF-D7** · CAPM, Alpha, Beta, and Hedging 21 questions · 7 groups
ML-D8 · Model evaluation 18 questions · 6 groups	**QF-D8** · Market Efficiency and Arbitrage 21 questions · 7 groups
ML-D9 · Generalization and complexity 18 questions · 6 groups	**QF-D9** · Financial Data and Research Bias 21 questions · 7 groups
ML-D10 · Choosing and deploying models 24 questions · 8 groups	**QF-D10** · Quantitative Strategy Design and Evaluation 21 questions · 7 groups

After the domain chapters: worked calculations, a 24-question mixed self-test, separate answers, and source-key cautions. Return here using “Contents” in the footer.

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A reliable way to answer

1. Decode the task before the statement

Most items use normal True/False. In 95 supplied questions, the stem explicitly reverses the convention: **True means the statement is incorrect**; False means it is accurate. A correct concept can therefore require the False label.

Concept	Controlling rule
Normal item	Evaluate the whole statement. Accurate → True; inaccurate → False.
Reversed item	Evaluate the whole statement. Inaccurate → True; accurate → False.
Nested negation	If an option says “the claim is false” or “a reviewer would be wrong to accept,” evaluate the embedded claim, then restore that negation.
Absolute wording	“Always,” “guaranteed,” “every,” and “zero effect” often overstate a tendency. Still read the scenario: an explicit mathematical assumption can make an exact claim valid.

2. Identify what is actually being compared

Separate prediction from trading profit; training error from generalization; raw return from risk-adjusted return; price from value; dollar neutrality from beta neutrality; and an estimated relationship from a guaranteed outcome.

3. Attach the calculation to its assumptions

Write the formula first, then substitute with consistent units. Use beginning-of-period weights, the stated fee base, the stated turnover convention, the correct risk denominator, and only information available at the decision date.

4. Retrieve, then repair

Read one domain, close the guide, and explain its group rules aloud. Work the mixed self-test without notes. For each miss, name the mistaken rule and redo one source question from that group. Finish with the key-caution pages so a defective label does not become a learned concept.

Three-pass review

Pass 1: concepts and contrasts. Pass 2: formulas and worked examples. Pass 3: mixed questions, wording, and weak groups. Spend more time where you cannot explain why an option is wrong, rather than rereading familiar pages.

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Formula sheet | Machine learning

Let N be observations, p features, k neighbors, e = prediction - target, and p_c a class proportion. These formulas summarize the distinctions tested in ML-D1 through ML-D10.

Concept	Controlling rule
Linear / polynomial	$\hat{y} = b_0 + \Sigma\, b_j\varphi_j(x)$. A polynomial is nonlinear in x but can remain linear in its fitted coefficients.
Squared-error fit	$SSE = \Sigma(y_i - \hat{y}_i)^2$. $R^2 = 1 - SSE/SST$, with $SST=\Sigma(y_i-\text{mean } y)^2$. Out-of-sample R^2 can be negative. Squared loss emphasizes large residuals.
Probability forecast	Logistic $p=1/(1+\exp(-\eta))$, with $\eta=b_0+\Sigma b_j x_j$. Binary Brier score = $\text{mean}[(p_i-y_i)^2]$. Numeric probabilities do not make classification into regression.
Regularization	Ridge: loss + $\lambda\Sigma b_j^2$. Lasso: loss + $\lambda\Sigma\lvert b_j\rvert$. Greater λ generally means less flexibility; intercept and scaling conventions matter.
Standardization	z = (x - μ_train)/s_train. Fit these statistics within each training fold; reuse them on its held-out data.
KNN regression	$\hat{y}(x)$ = mean(targets of k selected neighbors). Euclidean distance: $\text{sqrt}[\Sigma_j(x_j - z_j)^2]$. Scales and irrelevant dimensions affect neighbors.
Error metrics	MAE = mean|e|; MSE = mean(e^2); RMSE = sqrt[MSE]. RMSE ≥ MAE on the same errors. MSE uses target-units²; the others use target units.
Tree impurity	Gini = 1 - Σp_c^2; entropy = $-\Sigma p_c \ln(p_c)$; classification error = 1 - max(p_c). Pure nodes have Gini = entropy = 0.
Split improvement	Gain = I_parent - (n_L/N)I_L - (n_R/N)I_R. CART chooses the best admissible immediate gain, not the globally best future tree.
Tree limits	Depth d → at most 2^d leaves. min_samples_split constrains the parent; min_samples_leaf constrains each resulting leaf.
Bootstrap	N replacement draws from N rows omit a specified row with probability $(1 - 1/N)^N \approx 36.8\%$; about 63.2% of distinct original rows appear.
Ensemble intuition	For equal error variance σ^2 and common correlation ρ: Var(mean error) = $\sigma^2[\rho + (1 - \rho)/B]$. Shared error limits gains from more learners.

Pool anchors: ML-D4 (linear models), ML-D5 (KNN), ML-D6G1/G2/G4 (trees), ML-D7G1/G2 (bagging), ML-D8G2 (metrics). The ensemble-variance formula is a derived explanation of the pool's correlation argument.

Bias-variance: expected squared prediction error = bias² + variance + irreducible noise variance for a fixed process/information set. Residuals are often defined actual - predicted; squared/absolute losses are unchanged by reversing the sign.

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Formula sheet | Returns and risk

Concept	Controlling rule
Period / cumulative	$r_t = (P_t + \text{cash distribution} - P_{t-1})/P_{t-1}$. Linked $R = \Pi(1 + r_t) - 1$. Do not double-count distributions already in total-return data.
Mean / growth	Arithmetic mean = $\Sigma r/N$. Geometric mean = $[\Pi(1+r)]^{(1/N)} - 1$. CAGR = $(V_T/V_0)^{(1/\text{years})} - 1$; untreated external flows break the simple wealth comparison.
Annualization	Mean_ann ≈ m × mean_period; constant-rate effective return = $(1+r)^m - 1$; σ_ann ≈ sqrt(m)σ_period. The volatility rule needs appropriate dependence/stability assumptions.
Portfolio / borrowing	$r_p = \Sigma w_i r_i$ using beginning signed weights. One leveraged risky asset: r_equity = L r_asset - (L-1)r_borrow, before other costs.
Variance / covariance	Sample $s^2 = \Sigma(r\text{-mean})^2/(N-1)$; $\text{Cov}(A,B)=\rho_{AB}\sigma_A\sigma_B$. Use the question's variance convention if different.
Portfolio risk	$\sigma_p^2 = w^T\Sigma w$. Two assets: $w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\text{Cov}(A,B)$. Take the square root for volatility.
Sharpe	SR = mean(r_p-r_f)/std(r_p-r_f). Constant r_f permits std(r_p). Annualized periodic SR ≈ sqrt(m)SR under suitable assumptions.
Sortino	Sortino = [mean(r_p)-MAR]/downside deviation. A common downside convention is sqrt{mean[min(0,r-MAR)²]}; follow the supplied convention.
Information ratio	Active return a = r_p-r_b. Tracking error = std(a). IR = mean(a)/tracking error. Benchmark and risk-free rate are distinct references.
Drawdown / recovery	$DD_t = V_t/\text{running peak}_t - 1$; maximum drawdown magnitude = -min(DD). Recovery after fractional loss l is l/(1-l).
Weights / turnover	Net = Σsigned weights; gross = Σ|risky weights| when cash is excluded. One-way turnover may be ½Σ|Δw|; other items define total traded dollars/equity. Read the stem.

All rates and risk measures must share a horizon. Square-root annualization is not valid without qualification when returns are dependent. Zero denominator does not give an ordinary finite Sharpe or Sortino ratio.

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Formula sheet | Value, exposure, execution

Concept	Controlling rule
Present value	PV = FV/(1+r)^t. DCF = Σ CF_t/(1+r)^t + TV_T/(1+r)^T. Match the cash-flow claim, discount rate, and date.
Dividends / growth	No-growth value = D_1/r. Gordon growth = D_1/(r-g), requiring r>g and sustainable perpetual growth. If given D_0, use D_1=D_0(1+g).
Book / safety margin	Book equity = assets - liabilities; tangible book further excludes relevant intangible assets. MOS=(V-P)/V; modeled upside=(V-P)/P. Different denominators.
Scenario value	Expected value = Σp_sV_s; probabilities sum to 1. A weighted-average value does not remove the adverse scenario.
Beta	β_i = Cov(r_i,r_m)/Var(r_m) = ρ_im σ_i/σ_m. Correlation is bounded by ±1; beta need not be.
CAPM comparison	Expected r_i = r_f + β_i[E(r_m)-r_f]. Pool realized comparison: actual return - [r_f + β(r_m-r_f)]. One residual is not proof of persistent alpha.
Portfolio hedge	β_p=Σ$w_i\beta_i$. Beta dollars=Σsigned dollars$_i$ × β_i. Hedge short H=Lβ_L/β_H for positive hedge beta. Net dollars and beta dollars are different constraints.
Quotes / fills	Spread=ask-bid; midpoint=(ask+bid)/2. Fill VWAP=Σq_jP_j/Σq_j. A market buy consumes asks; a market sell consumes bids.
Short profit	Q(P_sell-P_cover) - dividends owed - borrow fees - other costs. Return divided by sale proceeds is not necessarily return on posted equity.
Execution shortfall	All intended shares filled: buy cost = Q(P_exec-P_decision)+fees; sell cost = Q(P_decision-P_exec)+fees. Positive means adverse cost.
Strategy economics	Expected P&L ≈ p_win×mean win - p_loss×mean loss - costs. Cost = traded dollars × cost rate. Hit rate alone cannot determine profit.

Quantitative finance pool anchors: QF-D5 (execution), D6 (valuation), D7 (beta/CAPM), D10 (strategy economics). Keep finance formulas educational and tied to the stated hypothetical scenarios.

ML-D1 / MACHINE LEARNING / 38 POOL QUESTIONS

Learning Paradigms & Problem Framing

Define the target, information available, decision, and operational constraints before choosing a learner; a great historical score alone does not make a good trading model.

Pool group / starting page	What to understand
G1 · Supervised vs. Unsupervised Learning ML-D1G1 · Q1-6 · p. 13	Supervised learning uses target labels; unsupervised learning finds structure without them. The same rows can support both stages. Naming a discovered cluster does not make clustering supervised or prove a useful regime.
G2 · Problem Type - Regression vs. Classification ML-D1G2 · Q1-8 · p. 17	Regression predicts magnitude; classification predicts a category. Numeric 0/1 labels remain categories, and a binary-event probability is normally probabilistic classification. Thresholding loses magnitude information; see the pool convention caution.
G3 · Training vs. Inference ML-D1G3 · Q1-6 · p. 23	Training estimates or updates the model; inference applies it. Cheap KNN fitting shifts work to prediction. Compare retraining window, query latency, throughput, and memory separately.
G4 · Hypothesis Space & Model Families ML-D1G4 · Q1-6 · p. 27	The hypothesis space determines representable relationships. Greater flexibility can reduce bias while increasing variance. Squared prediction error includes squared bias, variance, and irreducible noise under the usual setup.
G5 · Generalization vs. Memorization ML-D1G5 · Q1-6 · p. 32	Generalization means useful new-data performance, not memorization. Large datasets or extra epochs do not immunize flexible models. A small train/test gap can coexist with poor error everywhere.
G6 · What Defines a Good Model ML-D1G6 · Q1-6 · p. 36	A good model fits the decision objective and constraints. Accuracy, tail errors, stability, robustness, interpretation, latency, turnover, and net returns can conflict. A rare-event model can look accurate while missing every rare event.

Work it through · ML-D1G2Q7 · pool p. 21

Actual spread is -10 bps; forecasts -15 and -40 have squared errors 25 and 900 bps². Both correctly predict an inverted spread. Magnitude accuracy and category accuracy answer different questions.

Recall: a probability of a category is normally probabilistic classification; the pool contains a conflicting item. See source-key cautions.

ML-D2 / MACHINE LEARNING / 33 POOL QUESTIONS

Data & Sampling Discipline

A clean evaluation requires representative data and a pipeline that could have existed at the simulated decision time; both values and development choices can leak information.

Pool group / starting page	What to understand
G1 · Missing Data Handling ML-D2G1 · Q1-6 · p. 42	Investigate why data are missing. Train-fitted imputation, causal fill, missingness flags, and data-age flags serve different needs. Deletion may remove distressed cases; an imputed value is not observed truth.
G2 · Noise and Outliers ML-D2G2 · Q1-6 · p. 46	Verify extreme records against units, sources, and corporate actions. Correct confirmed errors but retain or explicitly model genuine tail events. Squared loss amplifies extreme residuals; clipping can hide risk.
G3 · Sampling Bias and Representativeness ML-D2G3 · Q1-6 · p. 51	Check inclusion, exclusion, and target-population coverage. More rows cannot restore missing bankrupt or acquired firms. Validation from the same selected sample does not establish transfer to a new market or regime.
G4 · Temporal Structure and Ordering ML-D2G4 · Q1-6 · p. 55	Train before evaluation; require matured labels and publication-time inputs. Overlapping windows share information. Sliding windows keep a fixed history length; expanding windows grow. Construct causal features before considering row-order permutations.
G5 · Train/Test Separation Discipline ML-D2G5 · Q1-6 · p. 60	Separate the whole fitted pipeline: imputation, scaling, PCA, feature selection, and model. Fit within each training partition. A later split does not undo full-data preprocessing or label-guided selection.
G6 · Data Snooping and Reuse ML-D2G6 · Q1-3 · p. 64	Searching many ideas creates lucky winners. Test-guided redesign turns the test into development data, even through shared scores. Logs improve transparency; only genuinely independent evaluation supplies fresh evidence.

Work it through · ML-D2G5Q4 · pool p. 61

Observed training values [1,2] have median 1.5. Adding future values [100,200] makes the median 51. Filling training gaps with 51 imports future information even without target labels.

Missingness: MCAR is unrelated to values; MAR is explained by observed data; MNAR still depends on missing values after conditioning on observed data.

ML-D3 / MACHINE LEARNING / 39 POOL QUESTIONS

Feature Engineering & Leakage

Useful features represent the right relationship using available information; scaling, selection, and transformations change model behavior without creating new raw information or guaranteeing generalization.

Pool group / starting page	What to understand
G1 · Feature Construction and Transformation ML-D3G1 · Q1-6 · p. 68	Ratios, growth, powers, and interactions can expose structure. Validate added capacity. Rolling estimation and ADF rejection do not establish structural stability.
G2 · Scaling and Normalization Effects ML-D3G2 · Q1-6 · p. 73	Fit scaling on training data. KNN neighbors depend on units; scaling does not equalize importance or remove a trend. Rolling z-scores can damp shocks as estimated volatility rises; ranks sacrifice magnitude.
G3 · Feature Relevance and Redundancy ML-D3G3 · Q1-6 · p. 78	Noise and redundancy add estimation risk. Collinearity can destabilize coefficients without always destabilizing predictions. Greedy selection can miss pure interactions; repeated searching can overfit.
G4 · Feature Selection Intuition ML-D3G4 · Q1-6 · p. 83	Feature reduction can lower variance, feed costs, and latency, but lose signal. Fit selection/PCA within each training fold. Ridge shrinks coefficients; lasso may zero them; PCA creates variance-based combinations, not target-optimal features.
G5 · Target Leakage ML-D3G5 · Q1-6 · p. 88	Audit feature lineage: post-outcome status, eventual last transactions, and full-data supervised selection leak outcomes. A contemporaneous peer feature is not automatically target leakage; availability and deployment design decide.
G6 · Look-Ahead Bias in Time-Series ML-D3G6 · Q1-6 · p. 93	Feature windows end by the decision time; use actual release/revision dates. Centered windows see the future. A final auction-price signal generally cannot claim an earlier order at that completed auction.
G7 · Too Good to Be True Features ML-D3G7 · Q1-3 · p. 97	Unexpected near-perfect results call for timing, lineage, and execution audits. Lag or remove suspect features and test untouched periods. Correlation or importance is not causal proof; leakage guarantees no particular accuracy.

Work it through · ML-D3G2Q5 · pool p. 76

Training mean 100 and SD 10 turn later values 120 and 150 into z=2 and z=5. Scaling has not removed the trend or bounded the output.

Recall: PCA preserves high feature variance, not necessarily predictive signal. Feature selection itself is part of the fitted and evaluated procedure.

ML-D4 / MACHINE LEARNING / 36 POOL QUESTIONS

Linear Models & Regression

Linear models are linear in coefficients: interpret conditional effects, inspect residual structure, and control capacity with valid representations and regularization.

Pool group / starting page	What to understand
G1 · Linear Model Structure and Interpretation ML-D4G1 · Q1-6 · p. 101	Interpret a coefficient with units and other included terms held fixed. A standardized feature moves one training SD per unit; all-zero standardized inputs represent training means. Historical coefficients are not causal or profit guarantees.
G2 · Assumptions of Linear Models ML-D4G2 · Q1-6 · p. 106	Check conditional mean, residual dependence/spread, omitted variables, and collinearity. OLS can be computed without normal errors; normality concerns some exact finite-sample inference. Overlapping targets reduce independent evidence.
G3 · Model Capacity and Polynomial Expansion ML-D4G3 · Q1-6 · p. 110	Powers/products enrich the feature basis while retaining linear-in-coefficients fitting. Select degree with validation. Correlated raw polynomial terms and high degree can destabilize fit and extrapolation.
G4 · Residuals and Model Fit ML-D4G4 · Q1-6 · p. 115	Residual = actual - predicted: positive means underprediction. OLS with an intercept has zero in-sample residual sum. Inspect curvature, changing spread, serial dependence, and tails; an average metric hides patterns.
G5 · When Linear Models Fail ML-D4G5 · Q1-6 · p. 120	Raw additive models miss U-shapes and conditional slope reversals. Powers or interactions can express them. Representation failure is not necessarily optimizer failure. Validate improvements on appropriate later data.
G6 · Regularization Intuition ML-D4G6 · Q1-6 · p. 126	Ridge penalizes squared coefficients; lasso absolute coefficients and can induce zeros. Scale sensibly; tune strength. Regularization may worsen training fit while helping validation, but too much underfits.

Work it through · ML-D4G1Q6 · pool p. 105

For $\hat{y}=.02+.045X_1-.012X_2-.038X_3$ at $X=(1,-1,.5)$, prediction=.02+.045+.012-.019=.058=5.8%. At all training-mean feature values, prediction is the 2% intercept.

Recall: changing variance is heteroskedasticity; time-linked errors are autocorrelation. Normal errors are not required to calculate OLS coefficients.

ML-D5 / MACHINE LEARNING / 36 POOL QUESTIONS

Instance-Based Learning (KNN)

KNN predicts from nearby stored examples, so feature geometry, local data quality, k, reference coverage, memory, and query latency determine success.

Pool group / starting page	What to understand
G1 · Distance & Similarity Behavior ML-D5G1 · Q1-6 · p. 133	Neighbors depend on representation, feature units, and distance metric. Train-fitted scaling can prevent large-unit features dominating. Euclidean, Manhattan, cosine, and rank-based comparisons need not choose the same neighbors.
G2 · Choice of k and Bias-Variance Tradeoff ML-D5G2 · Q1-6 · p. 139	Small k is locally flexible and noisy; large k smooths and can erase structure. At equal-weight k=N, regression predicts the global mean and classification the majority class. Select k by validation.
G3 · Sensitivity to Noise & Local Structure ML-D5G3 · Q1-6 · p. 144	A bad nearby label can dominate small k. More neighbors may dilute it but cannot guarantee robustness. Distinguish corrected errors from economically genuine extremes and evaluate relevant operating regimes.
G4 · High-Dimensional Effects ML-D5G4 · Q1-6 · p. 149	With fixed data, more dimensions make local coverage sparse and can concentrate distances. Standardizing irrelevant features does not make them useful. Training-only reduction may help but can remove target signal.
G5 · Computational Tradeoffs ML-D5G5 · Q1-6 · p. 154	Lazy fitting stores the reference data; ordinary KNN memory and brute-force distance work per query scale with Np. Indexes, prototypes, or approximate search trade resources against exactness.
G6 · Practical Use in Trading Systems ML-D5G6 · Q1-6 · p. 159	Measure actual N, p, query rate, memory, latency, and novel-state distance. KNN can suit slower analog or batch tasks; offline accuracy alone does not establish live feasibility.

Work it through · ML-D5G1Q2 · pool p. 134

Query (0,0), A=(3,0), B=(2,2): Euclidean distances 3 and sqrt(8)≈2.83 select B; Manhattan distances 3 and 4 select A. The metric changes the neighbor.

Recall: ordinary KNN retains N reference rows, not just the k neighbors returned for one query. The pool has a contradictory key letter.

ML-D6 / MACHINE LEARNING / 42 POOL QUESTIONS

Decision trees

A tree predicts with one path but learns its rules from many candidate partitions. The pool's standard CART chooses impurity-reducing splits; the recorded project tree uses correlation and a median. Keep these named algorithms separate.

Pool group / starting page	What to understand
G1 · Tree Structure and Prediction Flow ML-D6G1 · Q1-6 · p. 166	Follow one root-to-leaf path. A regression leaf returns its training-target mean; a classification leaf uses class information. Axis-aligned, constant leaves model thresholds but do not extrapolate a continuing target trend.
G2 · Splitting Criteria Intuition ML-D6G2 · Q1-6 · p. 173	CART greedily maximizes immediate weighted impurity reduction. Target labels guide splits; equal child sizes are unnecessary. Gini/entropy are zero at purity and maximal for a 50/50 binary mix.
G3 · Model Complexity and Overfitting ML-D6G3 · Q1-6 · p. 177	Deep or singleton-leaf trees can memorize noise: training error falls while validation worsens. Shallow trees can underfit interactions. Select capacity by validation, not perfect training fit.
G4 · Hyperparameters as Control Mechanisms ML-D6G4 · Q1-6 · p. 183	max_depth limits path length; min_samples_split limits parent eligibility; min_samples_leaf limits child size; max_leaf_nodes budgets terminal regions. Stronger limits trade finer structure for stability.
G5 · Stability and Sensitivity ML-D6G5 · Q1-6 · p. 188	Near-tied splits can change after tiny sample edits and reorganize descendants. Similar accuracy can hide unstable trades. Monotonic scaling does not fix sample sensitivity. See key cautions for G5Q5/Q6.
G6 · Strengths of Decision Trees ML-D6G6 · Q1-6 · p. 193	Trees capture conditional interactions and offer inspectable paths without mandatory feature standardization. Missing-value/category support is implementation-specific. Outlier robustness is qualified, not immunity.
G7 · Weaknesses and Failure Modes ML-D6G7 · Q1-6 · p. 199	Axis-aligned splits need a staircase for diagonal structure. Predictions can jump at thresholds; greedy choices and small noisy leaves can fail. A learned threshold is not a permanent causal market law.

Work it through · ML-D6G1Q2 · pool p. 167

Targets 1%, 2%, -1%, 2% give (1+2-1+2)/4 = 1%. Every new observation routed to that leaf gets 1%.

Recall: min_samples_split=10 can permit a 9+1 split. min_samples_leaf=10 cannot. Tree depth is not capped by the number of features.

ML-D7 / MACHINE LEARNING / 24 POOL QUESTIONS

Ensembles

Useful errors can partly cancel when learners differ. Shared bias, shared leakage, and shared regime blind spots survive averaging. Distinguish independent training fits from statistically independent prediction errors.

Pool group / starting page	What to understand
G1 · Motivation for Ensembles ML-D7G1 · Q1-6 · p. 206	Average regression outputs or vote among useful classifiers. Partly different errors can cancel; identical errors and shared bias remain. An ensemble is not guaranteed to beat its best member.
G2 · Bagging and Bootstrap Sampling ML-D7G2 · Q1-3 · p. 210	Bootstrap rows with replacement: duplicates and omissions are expected, labels unchanged. Fits can run independently, but overlapping data and dominant features keep errors correlated. Disjoint partitions are not usual bagging.
G3 · Random Forests ML-D7G3 · Q1-3 · p. 212	Random forests add random candidate features at each split. This trades tree strength against diversity. Using all features removes that component; fewer candidates do not guarantee better test error.
G4 · Diversity Among Learners ML-D7G4 · Q1-3 · p. 214	Judge diversity by held-out errors, not algorithm names. A weaker standalone model can add value through complementary mistakes. Random uninformative forecasts do not create useful diversity.
G5 · Boosting Intuition ML-D7G5 · Q1-3 · p. 216	Boosting fits sequential corrections or emphasizes poorly predicted cases. Small learning rate tempers each update; shallow learners can combine expressively. Noise/outliers may receive too much attention.
G6 · Bias vs. Variance in Ensembles ML-D7G6 · Q1-3 · p. 218	Bagging primarily stabilizes high-variance learners. Averaging identical underfit outputs preserves bias; boosting can add missing structure. Lower variance is not proof of unbiased predictions.
G7 · Failure Modes of Ensembles ML-D7G7 · Q1-3 · p. 220	Ensembling cannot legitimize leaked data. Many correlated learners can fail together in a new regime. Tune stage count, depth, and learning rate using time-aware validation; avoid best-training-fit selection.

Work it through · ML-D7G1Q2 · pool p. 207

Regression predictions 1%, 2%, 6% average to 3%. Class votes buy, buy, do not buy produce buy. Neither operation guarantees correctness.

Recall: ordinary out-of-bag evaluation can use later dates to predict earlier rows. Keep a time-aware final evaluation for forecasting.

ML-D8 / MACHINE LEARNING / 18 POOL QUESTIONS

Model evaluation

An error score is meaningful only relative to its units, benchmark, observations, and downstream decision. Protect final evaluation from every part of model selection, including preprocessing and repeated score inspection.

Pool group / starting page	What to understand
G1 · Role of Train, Validation, and Test Sets ML-D8G1 · Q1-3 · p. 224	Training fits parameters; validation chooses designs; untouched test supplies final evaluation. Repeated test feedback contaminates selection even without fit(). Refit selected settings on train+validation only before a still-untouched test.
G2 · Error Metrics - MAE, MSE, RMSE ML-D8G2 · Q1-3 · p. 226	MAE uses absolute error; MSE squares; RMSE square-roots MSE. Squared losses emphasize large misses. MSE has squared units; MAE/RMSE retain target units. RMSE is at least MAE for identical cases.
G3 · Baselines and Benchmarking ML-D8G3 · Q1-3 · p. 228	Compare with a training-mean regressor, simple class rule, or relevant investable strategy benchmark. A small-looking error or sophisticated method is not improvement by itself. Include costs and risk.
G4 · Fair Model Comparison ML-D8G4 · Q1-3 · p. 229	Compare the same held-out dates, metric, units, and information constraints. Fit scaling only on training. Calm versus crisis periods confound comparisons; percentage-point and decimal errors need conversion.
G5 · Metric Selection and Business Alignment ML-D8G5 · Q1-3 · p. 231	Connect accuracy to positions, move size, risk, turnover, and costs. Sign accuracy weights days equally; profit does not. Symmetric errors may miss asymmetric tail consequences. Keep statistical and economic diagnostics.
G6 · Variability and Stability of Results ML-D8G6 · Q1-3 · p. 233	Report typical performance and dispersion across windows/seeds, not just the best run. Equal mean error can hide crisis failures. Seed stability is different from stability across future regimes.

Work it through

For errors [-1, 1, 4], MAE=2, MSE=6, RMSE=sqrt(6)=2.449. All have no cancellation; only MAE/RMSE use target units.

Recall: train → choose on validation → freeze → final test. Higher directional accuracy does not guarantee higher net trading profit.

ML-D9 / MACHINE LEARNING / 18 POOL QUESTIONS

Generalization and complexity

Choose capacity using credible new-data performance. Bias, sample variance, irreducible noise, data relevance, and selection pressure explain why a perfect training fit can be a poor forecasting model.

Pool group / starting page	What to understand
G1 · Identifying Overfitting vs. Underfitting ML-D9G1 · Q1-3 · p. 237	High train and validation error suggests underfitting; low train but high validation error suggests overfitting. A small gap alone is insufficient. Prefer credible held-out performance over memorization.
G2 · Bias-Variance Tradeoff ML-D9G2 · Q1-3 · p. 239	Bias is systematic miss; variance is sensitivity across samples. More tree depth usually adds flexibility; larger KNN k smooths it. Choose expected unseen error, not minimum bias or variance in isolation.
G3 · Model Capacity vs. Data Size ML-D9G3 · Q1-3 · p. 241	New representative data can stabilize estimates; duplicates are not new evidence. More observations cannot repair an incapable model class. Old regimes may add rows without useful deployment information.
G4 · Cross-Validation and Generalization Estimation ML-D9G4 · Q1-3 · p. 242	Each CV round fits a new model and training-only preprocessing. For forecasting, train earlier and validate later. Model selection makes the winning CV score optimistic; retain a final untouched evaluation.
G5 · Detecting Overfitting in Practice ML-D9G5 · Q1-3 · p. 244	Audit spectacular results for leakage and selection effects. Perturbations reveal unstable trees. Searching hundreds of settings can overfit validation even when training results look ordinary. Record the search.
G6 · Controlling Model Complexity ML-D9G6 · Q1-3 · p. 246	Constrain depth/leaves, penalize large coefficients, or stop near the validation optimum. Too much restraint underfits. No control guarantees success, and final test feedback must not select the control.

Work it through · ML-D6G3Q3 · pool p. 178

Training error falls through depth 30, but validation reaches its minimum at depth 7 and rises thereafter. Depth around 7 is supported for selection; depth 30 is not favored merely for fitting training data best.

Recall: increasing tree depth adds flexibility; increasing KNN k removes local flexibility. A training-only fit can still be chosen using leaked test feedback.

ML-D10 / MACHINE LEARNING / 24 POOL QUESTIONS

Choosing and deploying models

Model choice is conditional on data shape, sample information, deployment constraints, and objective. A small validation advantage does not automatically dominate latency, stability, or explanation requirements.

Pool group / starting page	What to understand
G1 · Matching Models to Problem Characteristics ML-D10G1 · Q1-3 · p. 250	Match model shape, data, and objective. Linear baselines suit roughly linear effects; trees suit thresholds. Feature engineering can enrich a linear model. Validate rather than rank flexibility.
G2 · Accuracy vs. Interpretability Tradeoffs ML-D10G2 · Q1-3 · p. 252	Balance credible accuracy with explanation needs and practical significance. Deep trees can be hard to inspect. Internal models still need understood inputs and failure modes.
G3 · Training Time vs. Inference Time ML-D10G3 · Q1-3 · p. 254	Compare training with the retraining window and inference with query load. KNN defers work to prediction; linear inference is compact. Slow fitting need not imply slow serving.
G4 · Sensitivity to Noise and Data Quality ML-D10G4 · Q1-3 · p. 256	Correct confirmed defects. Squared loss amplifies target outliers; KNN depends on scale; trees are sample-sensitive. Bagging does not repair systematic data errors.
G5 · Data Size vs. Model Complexity ML-D10G5 · Q1-3 · p. 258	Limited samples cannot support unlimited adaptive choices. More representative data can help but increases KNN storage/query work. Old regimes and redundant indicators are not independent new information.
G6 · Real-Time Constraints in Trading Systems ML-D10G6 · Q1-3 · p. 260	Include feature computation, overhead, peak load, and tail latency. Missing a deadline can defeat better offline accuracy. Daily allocation and microsecond trading have different requirements.
G7 · Robustness and Stability in Production ML-D10G7 · Q1-3 · p. 262	Inspect regime errors, input perturbations, turnover, and exposures. Equal mean scores can hide instability. Monitoring and fallback can help; retraining more often is not automatically stabilizing.
G8 · Comparative Model Reasoning ML-D10G8 · Q1-3 · p. 263	Improve the system: engineer linear features, constrain or bag trees, or reduce KNN query cost. Re-evaluate tradeoffs; no algorithm is universally best.

Work it through · ML-D10G6Q1 · pool p. 260

An ensemble improves offline error by 0.2% but exceeds a 200-microsecond end-to-end budget. Better offline accuracy alone does not justify deployment; a faster adequate model can lead to better realized outcomes.

Recall: compare the whole decision pipeline under realistic load. An average score can hide costly regime, latency, or exposure failures.

QF-D1 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Investment Returns and Compounding

A return is a change divided by the capital exposed to that change. Each successive period starts from a new wealth level. Arithmetic averages describe average single-period observations; geometric averages describe realized compounded wealth. Always label the denominator, time horizon, cash-flow timing, and whether costs have been deducted.

Pool group / starting page	What to understand
G1 · Periodic and Cumulative Returns QF-D1G1 · Q1-3 · p. 272	Compute a period return relative to its beginning price. Link consecutive periods by multiplying growth factors. Equal dollar price changes need not mean equal percentage returns.
G2 · Compounding and Geometric Growth QF-D1G2 · Q1-3 · p. 274	CAGR is the constant annual rate connecting starting and ending wealth. Small recurring net-return differences accumulate through compounding. The Rule of 72 is a quick approximation.
G3 · Arithmetic vs. Geometric Average Returns QF-D1G3 · Q1-3 · p. 276	Arithmetic mean estimates a representative one-period outcome; geometric mean summarizes a sequence of positive gross returns. Geometric mean is at most arithmetic mean, with equality for identical periodic returns.
G4 · Annualization and Time-Scale Conversion QF-D1G4 · Q1-3 · p. 278	Put return and risk on matching horizons. Constant periodic return compounds; arithmetic mean annualization uses m times the period mean. Independent stable-variance returns permit volatility times square root of m.
G5 · Portfolio Return Aggregation QF-D1G5 · Q1-3 · p. 280	For one period with no intervening flow/rebalancing, use beginning signed weights: sum(weight × asset return). Include negative cash weight and borrowing cost when financed.
G6 · Gross, Net, and Benchmark-Relative Performance QF-D1G6 · Q1-3 · p. 282	Net performance deducts specified fees and costs. Active return subtracts the appropriate benchmark. Fee order and fee base determine the calculation.
G7 · Path Dependence and Return Sequencing QF-D1G7 · Q1-3 · p. 283	With no external flows, multiplying the same growth factors gives the same endpoint in any order; interim drawdown can still change. With withdrawals, order changes how much capital participates in later returns. A loss requires a larger percentage recovery.

Work it through · QF-D1G1Q1 · pool p. 272

Prices 50 → 55 → 52.25 give 55/50-1=10%, then 52.25/55-1=-5%. Linked return is 1.10×0.95-1=4.5%; $1 becomes $1.045.

Do not add equal gains/losses or assume equal percentage recovery. Same endpoint or average does not imply the same withdrawal experience or drawdown.

QF-D2 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Risk and Risk-Adjusted Performance

Choose the denominator that corresponds to the question. Sharpe asks about excess return versus risk-free return per unit of total volatility. Sortino focuses on shortfalls below a chosen target. Information ratio asks about active return versus a benchmark per unit of tracking error. None describes the complete loss distribution or chooses an investment for every investor.

Pool group / starting page	What to understand
G1 · Volatility, Variance, and Standard Deviation QF-D2G1 · Q1-3 · p. 287	Variance averages squared deviations; standard deviation is its square root. Equal means need not imply equal dispersion. Large observations strongly affect both mean and variance.
G2 · Downside Risk and Drawdown QF-D2G2 · Q1-3 · p. 289	Drawdown compares value with a previous running peak. Maximum drawdown is the worst historical peak-to-subsequent-trough decline; its depth and timing matter to withdrawals and constraints.
G3 · Sharpe Ratio QF-D2G3 · Q1-3 · p. 291	Sharpe = average excess return / standard deviation of excess returns, with compatible periods. For constant risk-free return the denominator equals portfolio-return standard deviation. Under IID assumptions annualized Sharpe is periodic Sharpe times sqrt(m).
G4 · Sortino Ratio and Downside-Focused Metrics QF-D2G4 · Q1-3 · p. 293	Sortino = mean return above minimum acceptable return (MAR) divided by downside deviation relative to that MAR. Only observations below the target contribute shortfall magnitudes.
G5 · Information Ratio and Active Risk QF-D2G5 · Q1-3 · p. 295	Tracking error is standard deviation of portfolio-minus-benchmark returns. Information ratio divides active return by that active risk.
G6 · Risk-Return Tradeoffs QF-D2G6 · Q1-3 · p. 297	Compare excess return, risk, horizon, spending needs, liquidity, and loss capacity. Metrics inform suitability; they do not erase constraints.
G7 · Limitations of Risk Metrics QF-D2G7 · Q1-3 · p. 299	Volatility is symmetric around the mean; distributions can share it but have different skewness, tails, and drawdown. Sample choice and best-of-many selection affect reported ratios.

Work it through · QF-D2G2Q1 · pool p. 289

A path peaks at 125 and drops to 90: drawdown=90/125-1=-28%. Recovery needs 125/90-1≈38.89%. Ending at 110 still leaves a 12% drawdown from the peak.

A precisely calculated metric on a selected calm window can mislead. Supplement with untouched testing, stress tests, downside metrics, and sensitivity analysis.

QF-D3 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Diversification and Correlation

Portfolio risk comes from weights, individual volatilities, and joint movement. The key question is what the candidate adds to the existing portfolio, not whether it looks good alone. Many names can share one economic exposure. Historical diversification can weaken in crises, so covariance estimates require uncertainty and scenario checks.

Pool group / starting page	What to understand
G1 · Correlation and Co-Movement QF-D3G1 · Q1-3 · p. 302	Correlation is standardized linear co-movement in [-1,+1]. Negative correlation is a tendency, not an every-period offset. Near zero means little linear association.
G2 · Covariance and Portfolio Risk QF-D3G2 · Q1-3 · p. 304	Covariance = correlation × both standard deviations. Portfolio variance includes squared-weight variance terms plus cross terms; square-root the result for volatility.
G3 · Specific vs. Systematic Risk QF-D3G3 · Q1-3 · p. 306	Issuer-specific risk can be diluted across genuinely different companies. Market, funding, liquidity, and other common exposures can remain.
G4 · Diversification Benefits QF-D3G4 · Q1-3 · p. 308	Imperfect co-movement can lower risk without proportionately lowering expected return. Judge defensive assets by their joint contribution. Perfect inverse co-movement can theoretically cancel volatility at suitable weights.
G5 · Concentration and Position Sizing QF-D3G5 · Q1-3 · p. 310	Weight determines the consequence of an error. Look beyond single-name caps to sector, factor, liquidity, and scenario concentrations. Risk contribution depends on covariance with the entire portfolio.
G6 · Diminishing Marginal Diversification QF-D3G6 · Q1-3 · p. 312	Marginal diversification usually diminishes as independent risks are already diversified and common risk remains. Redundant assets add little benefit.
G7 · Correlation Instability and Crisis Behavior QF-D3G7 · Q1-3 · p. 314	Stress can raise correlations through deleveraging and shared liquidity. Full-sample correlation can conceal high bad-state dependence. Optimization and leverage magnify errors.

Work it through · QF-D3G2Q1 · pool p. 304

Correlation 0.50 with volatilities 10% and 20% gives covariance 0.50×0.10×0.20=0.010. Doubling both volatilities makes covariance 0.040: four times larger, with correlation unchanged.

A stable past matrix is not a future guarantee. Use shrinkage, exposure limits, rolling estimates, and stress scenarios; instability does not make diversification worthless.

QF-D4 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Portfolio Optimization

An optimizer selects the best feasible weights for a stated objective and estimated inputs. Feasible, efficient, minimum variance, and maximum Sharpe are different concepts. Restrictions can lower the best modeled objective yet produce a more stable and implementable real strategy.

Pool group / starting page	What to understand
G1 · Portfolio Weights and Constraints QF-D4G1 · Q1-3 · p. 317	Fully invested means signed weights sum to one when all assets, including cash, are represented. Long-only separately requires nonnegative weights. Bounds, short rules, and gross exposure define the feasible set.
G2 · Minimum-Variance Portfolios QF-D4G2 · Q1-3 · p. 319	Global minimum variance minimizes risk over all feasible portfolios. Target-return minimum variance adds a return restriction. Covariance can make a combination safer than the safest asset alone.
G3 · Maximum-Sharpe Portfolios QF-D4G3 · Q1-3 · p. 321	Maximum Sharpe maximizes excess return per unit of volatility. With modeled risk-free borrowing/lending the tangency portfolio supplies the steepest capital allocation line; investor risk tolerance determines its mix with cash.
G4 · Efficient Frontier QF-D4G4 · Q1-3 · p. 323	A portfolio is dominated if another has at least as much return and no more risk, with one strict improvement. The efficient branch starts at GMV and extends toward higher returns.
G5 · Optimization Inputs and Estimation Error QF-D4G5 · Q1-3 · p. 325	Estimated means/covariances are uncertain inputs. Tiny apparent advantages can create extreme weights; an exact solver cannot repair wrong assumptions.
G6 · Constraints and Practical Portfolio Design QF-D4G6 · Q1-3 · p. 327	Leverage, position, sector, and turnover constraints encode implementability and can restrain noisy concentration. Binding limits can force gradual movement toward a target.
G7 · In-Sample Optimization and Out-of-Sample Performance QF-D4G7 · Q1-3 · p. 329	Test weights on later unused data; walk-forward estimates only from prior information. Hindsight annual optimal weights cannot be traded at the start of that same year.

Work it through · QF-D4G1Q2 · pool p. 318

Weights 1.20, 0.30, -0.50 sum to 1.00 net. Absolute weights sum to 2.00 gross. The portfolio is 100% net, 200% gross, and violates long-only despite summing to one.

Repeated test-guided revisions contaminate the test. Weight instability creates turnover costs. Equal weighting’s stability alone does not guarantee better returns.

QF-D5 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Market Microstructure and Trade Execution

The quote is a price for available liquidity, not a promise to fill arbitrary size. Execution trades off immediacy, price, and completion. Gross prediction quality matters only if the strategy can obtain positions at prices that leave a net edge.

Pool group / starting page	What to understand
G1 · Bid, Ask, Spread, and Midpoint QF-D5G1 · Q1-3 · p. 333	Bid is the highest available buy quote; ask the lowest sell quote. An aggressive buy pays ask, sell receives bid. Spread is ask-bid; midpoint averages them. Compare relative spread as well as dollars.
G2 · Market Orders and Limit Orders QF-D5G2 · Q1-3 · p. 334	Market orders prioritize execution with uncertain price. A buy limit caps price; sell limit sets a floor. A marketable limit can execute immediately up to the price limit and available depth.
G3 · Order Book and Market Depth QF-D5G3 · Q1-3 · p. 336	Market buys consume asks cheapest first; sells consume bids highest first. Compute quantity-weighted average fill across levels. Depth and spread are different liquidity dimensions.
G4 · Liquidity, Slippage, and Market Impact QF-D5G4 · Q1-3 · p. 338	Slippage is execution minus a stated benchmark with side-aware sign. Impact and limited depth can worsen fills. Larger participation in thin liquidity often increases cost.
G5 · Short Selling Mechanics QF-D5G5 · Q1-3 · p. 340	A stock short borrows and sells shares, later buys to return them. Profit from price decline is reduced by borrow fees, dividend equivalents, and other costs.
G6 · Transaction Costs and Implementation Shortfall QF-D5G6 · Q1-3 · p. 341	Implementation shortfall compares actual implementation against a defined decision-price portfolio, including stated costs. For fully filled simple examples, adverse price difference times quantity plus fees.
G7 · Execution Strategy and Order Splitting QF-D5G7 · Q1-3 · p. 343	Order splitting balances immediate impact against price movement, information leakage, and signal decay. Volume-sensitive execution responds to market activity; a fixed clock schedule does not guarantee VWAP.

Work it through · QF-D5G3Q1 · pool p. 336

Buy 500 shares at $10.01, 1,000 at $10.03, then 500 at $10.05. Total cost=$20,060; average fill=$10.03; final marginal fill=$10.05.

Slower is not always cheaper; greater urgency can rationally justify crossing spread. No schedule eliminates all impact, timing, and noncompletion risks.

QF-D6 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Valuation and Mispricing

Market price is observed; intrinsic value is conditional on forecasts and a model. The same stock can be below one defensible value estimate and above another. Estimating an attractive expected value is separate from surviving the path, financing the position, and realizing convergence.

Pool group / starting page	What to understand
G1 · Price vs. Value QF-D6G1 · Q1-3 · p. 348	Value depends on cash flows, growth, discount rates, scenarios, and probabilities. An uncertain price can represent probability-weighted outcomes.
G2 · Book Value and Asset-Based Valuation QF-D6G2 · Q1-3 · p. 349	Ordinary book equity=assets-liabilities; divide by shares for book/share. Tangible book additionally removes intangibles. Accounting value is not liquidation proceeds or all earning power.
G3 · Time Value of Money QF-D6G3 · Q1-3 · p. 351	Discount cash flows for timing and comparable risk; positive discounting makes an earlier identical payment more valuable. Compound invested cash using the matching periodic rate.
G4 · Discounted Cash Flow and Dividend Valuation QF-D6G4 · Q1-3 · p. 353	DCF adds discounted forecast cash flows and discounted terminal value. Constant-growth dividend valuation uses next period’s dividend divided by required return minus growth, with r>g.
G5 · Growth-Based Valuation QF-D6G5 · Q1-3 · p. 355	Grow cash flows multiplicatively. As perpetual growth approaches required return, the denominator shrinks and valuation sensitivity rises. Growth/value labels describe characteristics.
G6 · Margin of Safety and Scenario Analysis QF-D6G6 · Q1-3 · p. 357	Scenario values expose downside and estimation uncertainty. Probability-weighted value=sum(probability×value). Margin of safety uses discount relative to intrinsic value, not upside relative to purchase price.
G7 · Valuation-Based Long and Short Decisions QF-D6G7 · Q1-3 · p. 359	Compare upside, downside, catalysts, costs, risk limits, and horizon when turning valuation into a long or short.

Work it through · QF-D6G2Q1 · pool p. 349

Assets $500M minus liabilities $320M gives $180M book equity. With 30M shares, book/share=$6 and a $9 stock has P/B=1.5. A $60M write-down reduces book/share to $4.

Correct eventual value does not make an exited losing trade profitable. A short may be forced out, recalled, or rendered uneconomic by borrow cost before convergence.

QF-D7 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

CAPM, Alpha, Beta, and Hedging

Beta measures sensitivity to a chosen market model, not total risk or skill. Alpha is the intercept/return unexplained by the specified systematic exposure; a single residual is not persistent alpha. Hedge beta-weighted dollars, using signed positions and the stated equity denominator.

Pool group / starting page	What to understand
G1 · Market Model and Return Decomposition QF-D7G1 · Q1-3 · p. 363	Market model decomposes return into intercept + beta×market return + residual. An excess-return regression uses excess returns on both sides. The residual is actual minus modeled prediction.
G2 · Beta and Market Sensitivity QF-D7G2 · Q1-3 · p. 364	Beta=Cov(asset,market)/Var(market) is an estimated slope. A beta above1 magnifies modeled market response; negative beta reverses it.
G3 · Alpha and Active Performance QF-D7G3 · Q1-3 · p. 366	CAPM required return=r_f+beta×market excess. Compare actual to that risk-adjusted figure. Assess alpha’s standard error and implementation cost separately.
G4 · Correlation vs. Beta QF-D7G4 · Q1-3 · p. 368	Beta=rho×sigma_asset/sigma_market. Correlation measures standardized alignment; relative volatility changes slope.
G5 · Portfolio Beta and Exposure Aggregation QF-D7G5 · Q1-3 · p. 370	Portfolio beta sums signed weight×beta; dollar beta exposure sums signed value×beta and divides by equity for portfolio beta. Cash beta is0.
G6 · Dollar Neutrality vs. Beta Neutrality QF-D7G6 · Q1-3 · p. 371	Dollar-neutral means equal long and short dollars. Beta-neutral means signed beta-weighted dollars cancel. Different betas generally require unequal dollar notionals.
G7 · Hedging and Residual Risk QF-D7G7 · Q1-3 · p. 373	Hedge notional magnitude=long_notional×long_beta/hedge_beta for an opposing position. Modeled market return can cancel while asset-specific return remains.

Work it through · QF-D7G1Q1 · pool p. 363

In the stated raw-return market model, alpha 0.2%, beta 1.3, market return 2%, and residual -0.5% imply return 0.2%+1.3×2%-0.5%=2.3%. The market component is 2.6%.

Hedges leave residual/company, basis, liquidity, financing, and changing-beta risk. Beta-neutral and riskless are different claims.

QF-D8 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Market Efficiency and Arbitrage

First identify the information set and the kind of claimed advantage. Efficient markets concern persistent abnormal risk-adjusted returns after relevant costs, not the absence of price movement or positive returns. A pricing identity can create theoretical arbitrage; a historical relationship generally creates only a risky convergence thesis.

Pool group / starting page	What to understand
G1 · Weak-Form Market Efficiency QF-D8G1 · Q1-3 · p. 377	Weak-form efficiency concerns information in past prices and volume. Positive in-sample or gross returns from a historical pattern do not establish exploitable abnormal performance. A zero-mean random-walk innovation gives expected tomorrow price equal to today.
G2 · Semi-Strong Market Efficiency QF-D8G2 · Q1-3 · p. 379	Semi-strong efficiency includes public information and predicts rapid incorporation through informed trading, without requiring every investor to read it.
G3 · Strong-Form Market Efficiency QF-D8G3 · Q1-3 · p. 381	Strong-form efficiency extends to private information; private-information abnormal profits challenge it without necessarily contradicting semi-strong at that time. Public filings enter the public set when released.
G4 · Arbitrage and Law-of-One-Price Reasoning QF-D8G4 · Q1-3 · p. 383	Law of one price applies to identical state-contingent cash flows under stated frictionless, financing, and execution assumptions. Buy cheaper equivalent, sell dearer; evaluate net spread and lock both legs.
G5 · Statistical Arbitrage and Mean Reversion QF-D8G5 · Q1-3 · p. 385	A positive fitted spread z-score can motivate shorting the relatively expensive leg and buying the cheap leg under a mean-reversion thesis. Profit depends on relative movement.
G6 · Event-Driven and Relative-Value Strategies QF-D8G6 · Q1-3 · p. 387	Merger spreads price completion risk and time. Relative value expresses one asset outperforming another; both assets can fall while the long-short trade gains. Event reaction depends on surprise versus priced-in expectations.
G7 · Limits to Arbitrage and Market Efficiency QF-D8G7 · Q1-3 · p. 389	Mispricing can persist because shorting, capital, margin, liquidity, crowding, and timing constrain informed traders. Forced unwinds can amplify divergence.

Work it through · QF-D8G4Q1 · pool p. 383

Buy at $50.00 and simultaneously sell at $50.40. Gross spread=$0.40; subtract total cost $0.10 for $0.30 net per share under the stated executable, no-limit assumptions.

A correct eventual thesis can lose before convergence. More capital in the same trade is not always stabilizing. High borrow fees and recall can overwhelm the expected correction.

QF-D9 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Financial Data and Research Bias

Reconstruct what was knowable and tradable at the decision timestamp, then preserve independent evidence. A dataset can be accurate today yet invalid for a past decision. A pristine formula does not repair future information, a selected survivor universe, stale executable prices, or a repeatedly inspected holdout.

Pool group / starting page	What to understand
G1 · Temporal Ordering and Look-Ahead Bias QF-D9G1 · Q1-3 · p. 393	Use prior training data and later evaluation. Trailing windows avoid future observations; center windows include them. Forward labels can overlap test dates even when row dates do not.
G2 · Survivorship and Selection Bias QF-D9G2 · Q1-3 · p. 394	Point-in-time universes include historical constituents and dead securities/funds. Choosing markets or years after seeing outcomes creates selection bias.
G3 · Data Snooping and Multiple Testing QF-D9G3 · Q1-3 · p. 396	Many trials produce chance winners. Holdout-guided redesign makes the holdout part of development. Freeze specifications and test untouched later periods or markets with comparable structure.
G4 · Point-in-Time Data and Revisions QF-D9G4 · Q1-3 · p. 398	Use release timestamp and data vintage, not merely the economic period. Preserve original releases, revisions, and availability dates; add realistic processing delay.
G5 · Corporate Actions and Price Adjustments QF-D9G5 · Q1-3 · p. 400	Split-adjust prices and share counts consistently; include dividends for total returns and terminal delisting recovery for complete economic performance.
G6 · Missing, Stale, and Irregular Data QF-D9G6 · Q1-3 · p. 402	Missing, stale, and nonsynchronous data require economic interpretation. Forward filling creates mechanical zero returns and may hide nontradability; mean imputation is not a genuine observation.
G7 · Nonstationarity and Regime Change QF-D9G7 · Q1-3 · p. 403	Structural changes in businesses, leverage, regimes, competition, and market design can invalidate old relationships. Diagnose using rolling estimates, regime performance, and live data/execution diagnostics.

Work it through · QF-D9G3Q1 · pool p. 396

For 1,000 calibrated tests under a global null at level 0.05, expected false positives=1,000×0.05=50. Fifty nominal discoveries are not proof of fifty useful signals.

Longer old history does not guarantee a better current estimate. Remove an adverse period only with a substantive justification; retraining does not guarantee that any edge remains.

QF-D10 / QUANTITATIVE FINANCE / 21 POOL QUESTIONS

Quantitative Strategy Design and Evaluation

A deployable strategy is a chain: economic hypothesis → available data → forecast → risk/cost-aware target positions → executable trades → monitored net results. A good predictor can fail anywhere else in this chain. Validate the complete system against an investable, exposure-appropriate and simple alternative.

Pool group / starting page	What to understand
G1 · Investment Hypothesis and Signal Design QF-D10G1 · Q1-3 · p. 408	Specify a plausible mechanism, measurable signal, availability timestamp, and predictions that could falsify the hypothesis. Relevant controls and untouched validation help test alternatives.
G2 · Forecasts, Positions, and Trades QF-D10G2 · Q1-3 · p. 410	Forecast sign helps direction; magnitude, uncertainty, costs, covariance, and constraints determine whether/how much to trade. Use no-trade thresholds when expected edge is too small.
G3 · Backtesting and Benchmark Selection QF-D10G3 · Q1-3 · p. 411	Benchmark against an investable opportunity cost, matched exposures, and simple baselines. Freeze and test later; use walk-forward training only on earlier data.
G4 · Turnover, Costs, and Net Performance QF-D10G4 · Q1-3 · p. 413	Evaluate net return after spread, commissions, impact, borrow/financing, and specified fees. Turnover-aware/no-trade bands can improve economics independently of forecasting.
G5 · Capacity, Liquidity, and Scalability QF-D10G5 · Q1-3 · p. 415	Capacity depends on trade size relative to available liquidity, impact, speed, and competitors. More AUM can reduce percentage edge and eventually dollar profit.
G6 · Leverage, Exposure, and Risk Controls QF-D10G6 · Q1-3 · p. 417	Leverage magnifies gains and losses relative to equity. Net, gross, beta, sector, liquidity, and concentration exposures capture different risks. Stops can complement sizing/limits.
G7 · Robustness, Degradation, and Live Deployment QF-D10G7 · Q1-3 · p. 419	Monitor forecast quality, feature distributions, exposure, turnover, execution, order failures, data quality, and realized net performance. Use predefined review thresholds and staged rollout.

Work it through · QF-D10G2Q1 · pool p. 410

Forecast gross round-trip returns are +0.04%, +0.01%, and -0.03%; round-trip cost is 0.02%. The +0.01% long does not cover cost. Larger opportunities still need risk, confidence, and short-cost checks.

One bad week alone is weak retirement evidence; persistent multi-metric deterioration is stronger. Retraining cannot guarantee recovery; redesign or retirement may be appropriate.

CS7646 / EXAM 1

Worked calculations | ML

1. Linear features, conditional effect

Suppose $\hat{y} = 0.02 + 0.045X_1 - 0.012X_2 - 0.038X_3$ and standardized inputs are (1, -1, 0.5). Then $\hat{y} = 0.02 + 0.045 + 0.012 - 0.019 = 0.058$, or 5.8%. At all training-mean feature values, every standardized X is zero and the model predicts its 2% intercept.

ML-D4G1Q6 · pool p. 105

2. Distance is a modeling choice

Query (0,0), A=(3,0), B=(2,2). Euclidean distances: 3 and sqrt(8)=2.83, so B is closer. Manhattan distances: 3 and 4, so A is closer. Neither the algorithm name nor scaling alone defines “similar.” If selected neighbor targets are 1%, 2%, and 6%, their equal-weight forecast is 3%.

ML-D5G1Q2 · pool p. 134; last arithmetic is a derived drill.

3. Tree impurity and leaf prediction

A 50/50 binary node has Gini $1 - 0.5^2 - 0.5^2 = 0.5$. Split it into equal children with 90/10 and 10/90 proportions. Each Gini is 0.18; weighted child impurity is 0.18; gain is 0.32. A regression leaf with targets [1%, 2%, -1%, 2%] predicts 1%, regardless of the other leaves.

ML-D6G1Q6 · pool p. 171 (derived split values); ML-D6G1Q2 · pool p. 167

4. Units and penalties

For prediction errors [-1, 1, 4], MAE=(1+1+4)/3=2. MSE=(1+1+16)/3=6. RMSE=sqrt(6)=2.449. Changing the final error from 4 to 8 adds 4 units to absolute-error sum but 48 units² to squared-error sum. Squaring emphasizes big misses; it does not merely prevent cancellation.

Derived practice for ML-D8G2.

Explain the result, not just the arithmetic

A coefficient is conditional on the other included terms. A nearest neighbor depends on geometry. A split gain is local training improvement. A low metric is useful only against an appropriate baseline and decision objective.

CS7646 / EXAM 1

Worked calculations | Returns and risk

1. Compounding and order

Prices 50 → 55 → 52.25 give +10%, then -5%, and 1.10×0.95 - 1 = 4.5% cumulative. Separately, +30% then -20% gives 4% cumulative, a 5% arithmetic mean, and sqrt(1.04)-1 ≈ 1.98% geometric mean. With no external flows, reversing the order preserves final wealth; it can change drawdown.

QF-D1G1Q1 · pool p. 272; QF-D1G3Q1 · pool p. 276

2. Withdrawals make sequence matter

Start $1M and withdraw $100k after each year. +20%, then -20%: 1M×1.2-100k=1.1M; 1.1M×0.8-100k=$780k. Reverse the returns: 1M×0.8-100k=$700k; 700k×1.2-100k=$740k. The same return factors now act on different capital bases.

QF-D1G7Q2 · pool p. 284

3. Which risk denominator?

A earns 12% with 20% volatility; B earns 9% with 8% volatility; risk-free is 2%. Sharpe A=(12-2)/20=0.50; B=(9-2)/8=0.875. Lower raw return can have higher Sharpe. If another portfolio earns 10%, its benchmark 9%, tracking error 2%, risk-free 3%, and total volatility 12%, IR=0.50 while Sharpe≈0.58.

QF-D2G3Q1 · pool p. 291; QF-D2G5Q3 · pool p. 296

4. Covariance and diversification

Weights 0.6/0.4, variances 0.0225/0.0100, covariance 0.0030: variance = $0.6^2 \times 0.0225 + 0.4^2 \times 0.0100 + 2 \times 0.6 \times 0.4 \times 0.0030 = 0.01114$. Volatility = sqrt(0.01114) ≈ 10.55%. Do not report the variance as a percent volatility.

QF-D3G2Q2 · pool p. 305

Drawdown is path risk

From a prior peak of 125 to 90, drawdown is 90/125-1=-28%. Recovery requires 125/90-1≈38.89%, not 28%. A later ending value of 110 still sits 12% below that peak.
QF-D2G2Q1 · pool p. 289

CS7646 / EXAM 1

Worked calculations | Value and implementation

1. Forecast value versus purchase return

Scenario values $40/$60/$90 have probabilities 25%/50%/25%: expected value=$62.50. At price $50, margin of safety=(62.5-50)/62.5=20%; modeled upside=(62.5-50)/50=25%. The $40 bear case still loses 20%. If D_1=$3, r=9%, g=4%, Gordon value=$60; increasing r to 10% gives $50.

QF-D6G6Q1 · pool p. 357; QF-D6G4Q2 · pool p. 354

2. Hedge beta dollars, not just dollars

Long $30M at beta 1.5 creates $45M beta-dollar exposure. A hedge asset with beta 0.75 requires a $60M short to cancel it. Net dollars become -$30M: beta neutral, not dollar neutral. Separately, correlation 0.9, asset volatility 8%, and market volatility 16% imply beta 0.45, not 0.9.

QF-D7G6Q2 · pool p. 372; QF-D7G4Q1 · pool p. 368

3. Read the fee and turnover convention

With $100M capital, 15% gross profit, a 2% management fee, and incentive fee 20% of profit after management fees: incentive=.20×($15M-$2M)=$2.6M. Net=$10.4M. For a separate $10M strategy, 40% traded-dollar turnover at 0.10% cost trades $4M and costs $4,000; $3,000 gross profit becomes -$1,000.

QF-D1G6Q2 · pool p. 282; QF-D10G4Q2 · pool p. 414

4. A quoted price is not every fill

Buy 2,000 shares against asks 500 at $10.01, 1,000 at $10.03, then 500 at $10.05. Total cost=$20,060 and average fill=$10.03; marginal final fill=$10.05. A short of 500 shares at $20 covered at $18 earns $1,000 before a $200 dividend payment and $150 borrow fee, leaving $650.

QF-D5G3Q1 · pool p. 336; QF-D5G5Q2 · pool p. 340

One-way turnover is a different definition

Moving weights 50/50 → 20/80 gives ½(|-0.30|+|0.30|)=30% one-way turnover. A 10% one-way cap permits 40/60, not the full destination. QF-D4G6Q3 · pool p. 328

RETRIEVE FIRST / ANSWERS FOLLOW ALL QUESTIONS

Mixed self-test | 1-8

01 · ML-D1

A detector calls every one of 1,000 days normal; two were crash days. Give its accuracy and crash recall. Is its output quality established by accuracy?

02 · ML-D2

A dataset omits distressed firms with missing disclosures. Why can both deletion and neutral-value imputation mislead, and where should an imputer be fitted?

03 · ML-D3

You fit PCA and choose features on all dates, then split into chronological train/test periods. Is the later split enough? Does PCA guarantee preserving target signal?

04 · ML-D4

In $\hat{y}=2+3x_1-2x_2+4x_1x_2$, what is the effect of a one-unit rise in x_1 when $x_2=2$? Is this linear in its fitted coefficients?

05 · ML-D5

For query (0,0), compare A=(3,0) and B=(2,2) under Euclidean and Manhattan distance. What happens to equal-weight regression when k=N?

06 · ML-D6

A parent contains 10 observations. Is a 9+1 split allowed by min_samples_split=10 alone? What if min_samples_leaf=10? How many leaves can depth 3 permit?

07 · ML-D7

Distinguish bagging, random forests, and boosting in one sentence each. Why do 100 highly correlated learners not supply 100 independent opinions?

08 · ML-D8

Prediction errors are [-1,1,4]. Compute MAE, MSE, and RMSE, label units, and state which is most sensitive to large misses.

These are teaching drills derived from the pool, not additional official questions. For each answer, state the rule and assumptions before checking the solution.

RETRIEVE FIRST / ANSWERS FOLLOW ALL QUESTIONS

Mixed self-test | 9-16

09 · ML-D9

Training error decreases with depth. Validation error falls to depth 7, then rises. Which depth is supported, and may the test score select a replacement?

10 · ML-D10

An ensemble reduces offline error by 0.2% but misses a 200-microsecond trading deadline during bursts. Name the missing evaluation and a defensible decision.

11 · QF-D1

Prices move 50→55→52.25. Give each period return and cumulative return. Why is adding the two returns inaccurate?

12 · QF-D2

An asset earns 9%, risk-free is 2%, and volatility is 8%. Compute Sharpe. What replaces its reference and denominator in the information ratio?

13 · QF-D3

Two equally weighted assets each have 12% volatility and zero correlation. Find portfolio volatility. Does zero correlation guarantee they never fall together?

14 · QF-D4

Weights are 1.20, 0.30, -0.50. Find net and gross exposure. Does sum-to-one make the portfolio long-only? Can adding a restriction improve the same modeled maximum?

15 · QF-D5

Buy 2,000 shares: 500 at 10.01, 1,000 at 10.03, 500 at 10.05. Find average and final marginal fill. What does a buy limit order guarantee?

16 · QF-D6

Expected value is 62.50 and price is 50. Find margin of safety and modeled upside. Why are they different?

These are teaching drills derived from the pool, not additional official questions. For each answer, state the rule and assumptions before checking the solution.

RETRIEVE FIRST / ANSWERS FOLLOW ALL QUESTIONS

Mixed self-test | 17-24

17 · QF-D7

Long $30M at beta 1.5. The hedge asset has beta 0.75. Find the short needed for beta neutrality and whether it is dollar neutral.

18 · QF-D8

Which EMH form concerns all public information? Why is a historical pairs spread at z=2.5 not the same as a locked pricing arbitrage?

19 · QF-D9

A GDP value describes March but was first released in April and revised in June. Which version can a March strategy use? What is expected from 1,000 null tests at 5%?

20 · QF-D10

A $10M strategy trades 40% of portfolio value per day at 0.10% cost on traded dollars and earns $3,000 gross. Compute net profit and name a possible remedy.

21 · ML-D6

A binary parent is 50/50. Two equally sized children are 90/10 and 10/90. Compute Gini gain. Does the best gain establish the globally best tree?

22 · QF-D7

Correlation is 0.9; asset volatility is 8%; market volatility is 16%. Find beta and the modeled market component for a +2% market return.

23 · QF-D1

Start $1M. Withdraw $100k after each year. Compare +20%, then -20%, with the reverse order. Why do the endpoints differ?

24 · Reading protocol

In a reversed item, True means “incorrect.” The option says: “The claim that a single tree follows one path is false.” Which label follows? Explain both negations.

These are teaching drills derived from the pool, not additional official questions. For each answer, state the rule and assumptions before checking the solution.

ANSWER KEY / USE AFTER RETRIEVAL

Check your reasoning | 1-8

01 · ML-D1

Accuracy is 998/1,000=99.8%; crash recall is 0/2=0%. The relevant rare-event loss makes this a poor crash detector despite high accuracy.

ML-D1G6Q4 · pool p. 38

02 · ML-D2

Deletion can remove the cases of interest; neutral imputation can disguise their distress. Investigate the missingness mechanism and use indicators/sensitivity checks. Fit learned replacements only on eligible training data.

ML-D2G1Q1 · pool p. 42

03 · ML-D3

No. Full-data fitting influences representations and selection before the split. Refit them inside each training fold. PCA maximizes captured feature variance, not necessarily predictive target information.

ML-D3G3Q4 · pool p. 80

04 · ML-D4

The conditional slope is $3+4x_2=11$. It is nonlinear in the raw variables but linear in the fitted coefficients. The main-effect coefficient 3 alone is not the slope when $x_2=2$.

ML-D4G5Q1 · pool p. 120

05 · ML-D5

Euclidean selects B: sqrt(8)<3. Manhattan selects A: 3<4. At k=N, every query receives the training-target mean and local feature structure is lost.

ML-D5G1Q2 · pool p. 134

06 · ML-D6

The 9+1 split is allowed by the parent-only split threshold. It is forbidden by minimum leaf size 10. Depth 3 permits at most $2^3=8$ leaves, not necessarily exactly eight.

ML-D6G4Q6 · pool p. 187

07 · ML-D7

Bagging independently fits replacement samples and aggregates. Random forests add random candidate features per split. Boosting fits sequential corrections. Shared data, predictors, bias, and regime failures correlate errors, limiting averaging gains.

ML-D7G6Q3 · pool p. 219

08 · ML-D8

MAE=2 target units; MSE=6 target units^2; RMSE=sqrt(6)≈2.449 target units. MSE and RMSE emphasize large errors through squaring. RMSE is not more outlier-robust than MAE.

ML-D8G2Q2 · pool p. 226

ANSWER KEY / USE AFTER RETRIEVAL

Check your reasoning | 9-16

09 · ML-D9

Depth near 7 is supported by validation. Later complexity appears to overfit. The final test must not select a replacement; test-guided revisions consume its independence.

ML-D6G3Q3 · pool p. 178

10 · ML-D10

Measure end-to-end feature, model, and system latency under realistic load, including tail latency. The ensemble may be unusable; a faster adequate model can create better realized outcomes.

ML-D10G6Q1 · pool p. 260

11 · QF-D1

The returns are +10% and -5%; compounded return is 1.10×0.95-1=4.5%. The second percentage acts on 55, so simple addition uses inconsistent capital bases.

QF-D1G1Q1 · pool p. 272

12 · QF-D2

Sharpe=(9-2)/8=0.875. Information ratio uses portfolio-minus-benchmark active return divided by tracking error, the standard deviation of that active-return series.

QF-D2G3Q1 · pool p. 291

13 · QF-D3

Variance=$0.25\times0.12^2+0.25\times0.12^2=0.0072$; volatility≈8.49%. Zero linear correlation neither implies independence nor prevents simultaneous losses.

QF-D3G4Q1 · pool p. 308

14 · QF-D4

Net=1.00; gross=2.00. A negative holding violates long-only despite net 100%. Restricting an unchanged feasible optimization cannot improve its mathematical maximum, though realized robustness may improve.

QF-D4G1Q2 · pool p. 318

15 · QF-D5

Average fill is $20,060/2,000=$10.03; marginal final fill=$10.05. A buy limit constrains the acceptable purchase price, not whether or when the order fills.

QF-D5G3Q1 · pool p. 336

16 · QF-D6

Margin of safety=(62.50-50)/62.50=20%; upside=(62.50-50)/50=25%. One uses value as its denominator, the other invested price. Neither guarantees convergence.

QF-D6G6Q1 · pool p. 357

ANSWER KEY / USE AFTER RETRIEVAL

Check your reasoning | 17-24

17 · QF-D7

Beta dollars are $45M; short=45/0.75=$60M. Beta cancels but net dollars are $30M-$60M=-$30M. Other risks, costs, and beta-estimation errors remain.

QF-D7G6Q2 · pool p. 372

18 · QF-D8

Semi-strong efficiency concerns all public information. A high z-score measures historical unusualness; it does not guarantee convergence or preserve the relationship. Statistical arbitrage retains price, financing, timing, and model risk.

QF-D8G5Q1 · pool p. 385

19 · QF-D9

The March strategy cannot use a not-yet-released April value or June revision. Use the latest vintage actually available by the decision time. At a calibrated 5% null-test level, expected false positives are 1,000×0.05=50.

QF-D9G3Q1 · pool p. 396

20 · QF-D10

Traded dollars=$4M; costs=$4,000; net=-$1,000. Cost-aware no-trade bands, lower turnover, or better execution may improve economics, but must be tested with realistic costs and opportunity tradeoffs.

QF-D10G4Q2 · pool p. 414

21 · ML-D6

Parent Gini=0.5; each child=0.18; weighted gain=0.32. This establishes only the best current admissible split under the criterion, not a globally optimal full tree or minimum test error.

ML-D6G1Q6 · pool p. 171

22 · QF-D7

Beta=0.9×8/16=0.45. Market component=0.45×2%=0.9%, before intercept and residual. High correlation is not the same as high beta.

QF-D7G4Q1 · pool p. 368

23 · QF-D1

First sequence: $1.1M after year-one withdrawal, then $780k. Reverse: $700k, then $740k. Withdrawals change the capital to which the later factor applies; return order now affects ending wealth.

QF-D1G7Q2 · pool p. 284

24 · Reading protocol

The embedded claim is accurate; the option denies it, so the complete option is incorrect. Under reversed labeling, select True. Evaluate the sentence first, then apply the question’s label convention.

ML-D6G1Q1 · pool p. 166

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Source-key cautions | Clear conflicts

The table distinguishes the supplied key from the concept. Keep these separate when practicing the exact source. These are identified source inconsistencies, not silent corrections to the published pool.

Concept	Controlling rule
ML-D5G5Q4 E **Pool p. 156**	**Published: True.** It says KNN stores only k observations for future predictions. **Concept: False.** Ordinary KNN retains N reference rows; k selects neighbors for a particular query. Its own explanation discusses the larger memory requirement.
ML-D6G5Q5 B **Pool p. 191**	**Published: True.** It says a monotonic transform changes observation order and stabilizes the tree. **Concept: False.** A strictly increasing transform preserves order and ordinarily preserves candidate partitions. The printed B explanation discusses option C.
ML-D6G5Q5 D **Pool p. 191**	**Published: False.** It correctly calls tree variance a motivation for bagging/random forests. **Concept: True.** Printed explanations B/C/D appear permuted; D instead discusses the transform claim.
ML-D6G5Q6 B **Pool p. 192**	**Published: False.** The statement correctly says monotonic transformations preserve partitions and do not inherently stabilize changed samples. **Concept: True.** Its explanation affirms the statement; there is no reversed-label instruction.
ML-D1G2Q8 A/D **Pool p. 22**	**Published: True / True.** This item treats a recession probability as regression and changing to category output as a task change. **Concept:** predicting the probability of a binary category is normally probabilistic classification. D1G2Q2 rejects “numeric output therefore regression.”
ML-D3G5Q6 A/C **Pool p. 92**	**Published: False / False in a reversed item.** The stem says one signal is clean, but the malformed Alpha statement and key are inconsistent. The rationale intends Alpha leaky and point-in-time Gamma clean. Cross-sectional availability still needs context.

How to use this page

Learn the defensible rule. When practicing a flagged exact option, note its published label separately. A mismatch should trigger review of the source wording, not memorization of a false general principle.

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Source-key cautions | Important qualifications

Published labels below refer to the exact option, including reversed stems. The teaching rule narrows overbroad wording without changing the source record.

Concept	Controlling rule
ML-D3G3Q4 E **Pool p. 80**	**Published True:** PCA can remove redundancy “without discarding” signal. **Qualification:** leading variance components may discard predictive information; fit and validate the reduction within training.
ML-D3G3Q5 C **Pool p. 81**	**Published False, reversed:** the option says collinearity “will” destabilize predictions. **Qualification:** coefficient instability need not make predictions unstable when the joint feature distribution remains similar.
ML-D3G3Q6 A **Pool p. 82**	**Published False, reversed:** it calls greedy forward selection fundamentally unable to capture interactions. **Qualification:** it can miss pure joint effects with no standalone gain; the universal claim is too strong.
ML-D3G5Q4 **Pool p. 90**	**Published selects B/D/E and labels A False.** Cross-sectional peer summaries are called leakage. **Qualification:** whether they leak depends on contemporaneous availability, target use, and the intended deployment population.
ML-D2G4Q5 B/E **Pool p. 58**	**Published False / True.** The pool rejects all within-training shuffling and mixes sliding with expanding windows. **Qualification:** causal feature construction and sequence-dependent learners need order; permuting already formed tabular pairs may be harmless.
ML-D2G1Q6 C **Pool p. 45**	**Published False, reversed:** a full training-period mean is called look-ahead. **Qualification:** it is valid for later held-out prediction, but not for simulating earlier decisions before its input data existed.
ML-D2G3Q4 C **Pool p. 53**	**Source wording mismatch:** the statement discusses including acquired stocks, while the explanation discusses their exclusion. Survivorship bias comes from outcome-related selection; missing acquired winners can also matter.

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Source-key cautions | Avoid false absolutes

Concept	Controlling rule
Probabilities and losses ML-D1G2Q4 p. 19; Q8 p. 22	A squared probability error is the Brier score for binary outcomes; it is not intrinsically invalid because the task is classification. Accuracy evaluates thresholded decisions, not the full quality of unthresholded probabilities.
OLS and noise ML-D3G1Q4 p. 70; D4G4Q6 p. 119	OLS is computable without stationary inputs or normal errors. Drift can invalidate prediction; heavy tails can invalidate exact normal-theory finite-sample inference. Do not infer that every coefficient or standard-error calculation necessarily fails.
Bias/variance shorthand ML-D1G4Q6 p. 31; D1G5Q4/Q5 p. 34	ε is a random error; its conditional variance contributes irreducible squared error. A U-shaped curve, narrowing gap, or “entropy matching” is not a universal operational definition of generalization. Judge relevant held-out risk.
KNN and compute ML-D5G2Q4 p. 141; D5G5Q4 p. 156; D5G6Q6 p. 163	1-NN interpolation assumes self-neighbors and no conflicting duplicate labels. Larger k smooths; it need not produce a linear boundary. Lazy fitting still incurs storage/preprocessing cost. Hardware timing is measured, not guaranteed by an algorithm name.
Trees and MAE ML-D6G6Q5 p. 197; D8G2Q1 p. 226	Monotonic transforms ordinarily preserve training partitions, not numeric threshold values or every implementation outcome. MAE’s “equal treatment” means equal cost per unit error, not identical loss for different magnitudes.
OOB and causality ML-D7G1Q6 p. 209; D3G6Q4/Q5 p. 95	OOB omission is not past-to-future validation. Centered windows leak but do not imply a universal 90% accuracy. Trading later fixes signal timing only if the claimed later execution is feasible; realized VWAP is not a free guaranteed fill.
Quant finance wording QF-D6G3Q2 C p. 352; D9G6Q1 D p. 402	Later cash flows are more sensitive in relative PV terms; an absolute dollar PV gap need not widen for every rate. Stale fills distort volatility and dependence, but autocorrelation does not have a universal direction of change.
Arbitrage and concentration QF-D8G4Q1 D p. 383; D3G5Q2 A p. 311	The arbitrage option has a malformed “not completed” clause and published False; locking both legs reduces leg risk. Equal name weights reduce imbalance, but do not erase issuer exposure or shared sector concentration.

The app retains the broader audit beside exact source items. These pages prioritize conflicts and qualifiers most likely to cause a wrong general rule. Nested-negation items such as QF-D6G7Q2 B/C are logically consistent: evaluate the wrapper as well as the arithmetic.

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Coverage, sources, and final checks

Authoritative scope

ML4T Exam Question Pool, edition Rev081020261230. This guide covers its Exam 1 / Part 1 machine-learning and quantitative-finance sections: 518 question IDs, 135 groups, and 20 domains. Counts include source duplicates. Every group is mapped in its domain chapter; the mixed self-test is deliberately smaller and teaches transfer rather than replicating the full key.

What changed from the earlier course-wide review

This version concentrates on the supplied Exam 1 pool. It expands CART, regularization, feature selection/PCA, missingness, probability framing, error metrics, production constraints, compounding, downside risk, tracking error, growth valuation, execution costs, and research bias. Reinforcement learning, Q-learning, Dyna, and options are outside this Exam 1 synthesis.

How the course evidence is used

The earlier lecture-based review and downloaded subtitles support core mechanisms and examples. The pool controls the present exam scope and scenario conventions. Specific distinctions are preserved: standard CART versus the recorded correlation/median project tree; an N-draw bootstrap versus the lecture's 60%-size example; benchmark-active information ratio versus residual-return discussion; and fee bases stated in each problem.

Source files

Question pool: ML4T Exam Question Pool.pdf
Earlier review: CS7646_Exam_Review.md
Lecture folder: CS7646_Lectures
Reference syntax: ML-D6G4Q6 means ML domain 6, group 4, question 6. "Pool p." means its 1-based PDF start page; a long question may continue onto later pages.

Further checks used for technical conventions

Official scikit-learn and statsmodels documentation support narrow clarifications on probability classification, scoring, PCA, preprocessing leakage, KNN resources, and OLS diagnostics. They do not override the supplied labels. See the interactive app's source audit for the exact-item discussion.

scikit-learn: model evaluation · common pitfalls · PCA · statsmodels: regression diagnostics

Ready when you can do all four

1. Explain each domain's rules without looking.
2. Reproduce formulas with units and assumptions.
3. Diagnose why plausible distractors fail.
4. Resolve normal/reversed labels and identify the flagged source inconsistencies.